

Section : Physics

Q.1 The refractive indices of glass and water are $3/2$ and $4/3$, respectively. If light passes from glass to water, the critical angle is

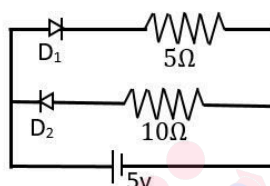
1. $\sin^{-1}(8/9)$
2. $\sin^{-1}(7/8)$
3. $\sin^{-1}(10/9)$
4. $\sin^{-1}(9/8)$

Options 1. 1

2. 2
3. 3
4. 4

Q.2

In the given circuit, D_1 and D_2 are normal diodes. The current drawn from the battery by the diode network is



1. 0.33 A
2. 0.86 A
3. 1.0 A
4. 0.5 A

Options 1. 1

2. 2
3. 3
4. 4

Q.3 In a series LCR circuit $X_L = X_C$. Which of the following is correct ?

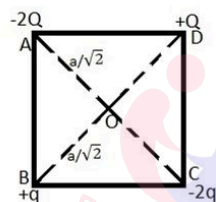
1. Current leads voltage in phase
2. Current lags voltage in phase
3. The current amplitude is maximum
4. Current amplitude is minimum

Options 1. 1

2. 2
3. 3
4. 4

Q.4

Four point charges $-2Q$, $+q$, $-2q$ and $+Q$ are placed at four corners of a square A, B, C and D, respectively, as shown in the figure and $OA = OB = a/\sqrt{2}$. If the potential at the center of a square is zero, what is the relation between Q and q ?



1. $Q = q$
2. $Q = -q$
3. $Q = 2q$
4. $Q = -2q$

Options 1. 1

2. 2
3. 3
4. 4

Q.5

A coil has an inductance of $L = 0.02$ H and the current is increasing at the rate of $\frac{dI}{dt} = 200$ A/s.

The emf induced in it, ε , is:

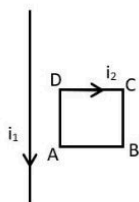
1. -2 V
2. -4 V
3. $+4\text{ V}$
4. $+6\text{ V}$

Options

- 1.
- 2.
- 3.
- 4.

Q.6

In the given diagram, the direction of current is given. Due to the presence of the current carrying wire, the loop ABCD will be



1. stationary
2. attracted towards the wire
3. repelled by the wire
4. rotating around the wire

Options

- 1.
- 2.
- 3.
- 4.

Q.7

A pair of adjacent coils has a mutual inductance of 2.0 H . If the current in one coil changes from 0 to 10 A in 0.5 s , the change of the flux linkage with the other coil is

1. 10 Wb
2. 20 Wb
3. 30 Wb
4. 40 Wb

Options

- 1.
- 2.
- 3.
- 4.

Q.8

An element $\Delta\vec{l} = 0.5\hat{j}\text{ cm}$ is placed at the origin and carries a large current $I = 10\text{ A}$. The magnetic field at point $(50\text{ cm}, 0)$ is

1. $2 \times 10^{-8}(\hat{i})\text{ T}$
2. $2 \times 10^{-8}(-\hat{i})\text{ T}$
3. $2 \times 10^{-8}(\hat{k})\text{ T}$
4. $2 \times 10^{-8}(-\hat{k})\text{ T}$

Options

- 1.
- 2.
- 3.
- 4.

Q.9

The displacements of four light waves are given below. If the amplitudes a_1 and a_2 are nearly equal, then the interference fringes may be observed due to superposition of:

- (A) $y = a_1 \sin \omega t$
- (B) $y = a_2 \sin(\omega t + \phi)$
- (C) $y = a_1 \sin 2\omega t$
- (D) $y = a_2 \sin 3(\omega t + \phi)$

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (A) and (C) only
3. (B) and (C) only
4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.10 An emf $E = 200 \sin 300t$ volt is applied across an inductor coil having a resistance of 1.0Ω . If the maximum current is found to be 10 A, then the value of inductance is approximately:

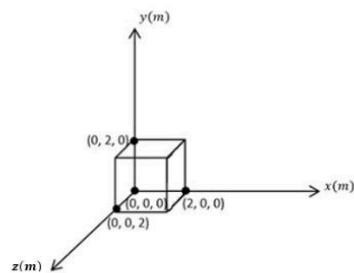
1. 23 mH
2. 41 mH
3. 67 mH
4. 53 mH

Options 1. 1

2. 2
3. 3
4. 4

Q.11

A cube is placed at the origin as shown in the figure. The electric field in the region is given as $\vec{E} = (200x^2\hat{i}) \frac{\text{N}}{\text{C}}$. The flux coming out of the cube is:



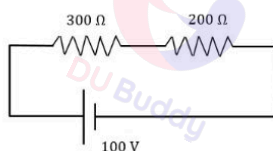
1. $3200 \frac{\text{Nm}^2}{\text{C}}$
2. $0 \frac{\text{Nm}^2}{\text{C}}$
3. $800 \frac{\text{Nm}^2}{\text{C}}$
4. $1600 \frac{\text{Nm}^2}{\text{C}}$

Options 1. 1

2. 2
3. 3
4. 4

Q.12

A voltmeter of resistance 600Ω is used to measure the potential drop across 300Ω resistor. The measured value of potential drop will be



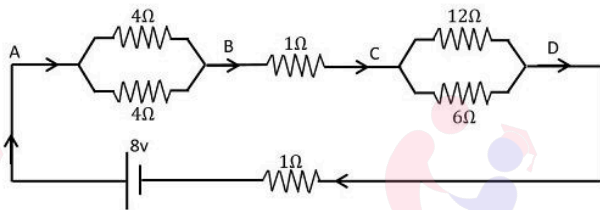
1. 50 V
2. 100 V
3. 60 V
4. 10 V

Options 1. 1

2. 2
3. 3
4. 4

Q.13

A network of resistors is connected as shown in the figure. The potential drop across 4Ω resistors is



1. 2 V
2. 4 V
3. 1.5 V
4. 2.5 V

Options

1. 1
2. 2
3. 3
4. 4

Q.14 Two nuclei have mass numbers 125 and 27, respectively. The mass density ratio of their nuclei will be

1. 5:3
2. 125:27
3. $\sqrt[3]{125} : \sqrt[3]{27}$
4. 1:1

Options

1. 1
2. 2
3. 3
4. 4

Q.15 The relation below correctly represents the frequency of light emitted when an electron in a hydrogen atom jumps from a higher (n_i) energy level to (n_f) a lower energy level: $\nu = Rc \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$

Predict the values of n_i , n_f , respectively for the Paschen series, which will give the highest wavelength in the series

1. 3, 2
2. 4, 3
3. 5, 4
4. ∞ , 3

Options

1. 1
2. 2
3. 3
4. 4

Q.16 If μ_0 is the permeability and ϵ_0 is the permittivity of free space, then the velocity of light (c) is:

1. $\sqrt{\frac{\mu_0}{\epsilon_0}}$
2. $\frac{1}{\sqrt{\mu_0 \epsilon_0}}$
3. $\sqrt{\frac{\epsilon_0}{\mu_0}}$
4. $\sqrt{\mu_0 \epsilon_0}$

Options

1. 1
2. 2
3. 3
4. 4

Q.17 A thin equiconvex lens of radius of curvature R made up of glass ($\mu_g = 1.5$) is dipped in water ($\mu_w = 1.3$). Its focal length in water is:

1. $3.25R$
2. $6.25R$
3. $R/2$

4. 4.25R

Options 1. 1

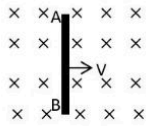
2. 2

3. 3

4. 4

Q.18

A rod AB moves with uniform velocity v in a uniform magnetic field as shown in the figure. Identify the correct statement(s) among the following.



(A) Rod becomes electrically charged.

(B) End A becomes positively charged.

(C) End B becomes positively charged.

(D) Rod becomes hot due to joule's heating.

Choose the **correct** answer from the options given below:

1. (B) only

2. (A) and (D) only

3. (B) and (D) only

4. (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.19

Three capacitors of capacitances $2\ \mu\text{F}$, $3\ \mu\text{F}$ and $6\ \mu\text{F}$ are connected in series. The combination is connected to a 12 V supply. The respective potential differences across them will be

1. 3 V, 4 V and 5 V

2. 2 V, 4 V and 6 V

3. 6 V, 4 V and 2 V

4. 4 V across each capacitor

Options 1. 1

2. 2

3. 3

4. 4

Q.20

A current I flows in a circular loop of radius R . The magnetic field at a point on its axis at distance r from the center of the loop is

1. $\frac{\mu_0 I}{2\sqrt{2}r}$

2. $\frac{\mu_0 I}{4\sqrt{2}r}$

3. $\frac{\mu_0 I}{8\sqrt{2}\pi r}$

4. $\frac{\mu_0 I}{316\sqrt{2}\pi r}$

Options 1. 1

2. 2

3. 3

4. 4

Q.21

The radius of curvature of the path of a charged particle in a uniform magnetic field is directly proportional to the

1. charge of the particle

2. momentum of the particle

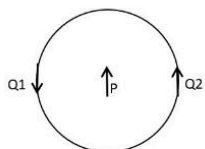
3. kinetic energy of the particle
4. flux density of the field

Options 1. 1

2. 2
3. 3
4. 4

Q.22

The given figure shows two magnetized needles and a magnet P with their magnetic moments, shown with the arrow sign. Which of the following statements are correct in respect of this configuration?



- (A) Q_1 is in stable equilibrium
(B) Q_2 is in unstable equilibrium
(C) Q_1 is in unstable equilibrium
(D) Q_2 is in stable equilibrium

Choose the **correct** answer from the options given below:

1. (A) and (B) only
2. (A) only (D) only
3. (B) and (C) only
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.23 Arrange the following electromagnetic waves in the increasing order of their energy.

- (A) Ultraviolet rays
(B) Infrared rays
(C) Microwaves
(D) X-rays
(E) Gamma rays

Choose the correct answer from the options given below:

1. (A), (B), (C), (D), (E)
2. (E), (D), (C), (B), (A)
3. (C), (B), (A), (D), (E)
4. (B), (A), (D), (E), (C)

Options 1. 1

2. 2
3. 3
4. 4

Q.24 Three resistances of $1\ \Omega$, $2\ \Omega$ and $3\ \Omega$ are given. In order to obtain an equivalent resistance of $11/3\ \Omega$ they are to be connected as

1. $3\ \Omega$ in series with parallel combination of $1\ \Omega$ and $2\ \Omega$
2. $3\ \Omega$ in parallel with series combination of $1\ \Omega$ and $2\ \Omega$
3. all in parallel
4. $1\ \Omega$ in series with parallel combination of $2\ \Omega$ and $3\ \Omega$

Options 1. 1

2. 2
3. 3
4. 4

Q.25 If 2000 electric field lines enter a given volume of closed space and 4000 field lines diverge from it, then the total charge within the volume is

(Given $\epsilon_0 = 8.85 \times 10^{-12}\ \text{C}^2\text{m}^{-2}\text{N}$)

1. $1.01 \times 10^{-6}\ \text{C}$
2. $1.77 \times 10^{-8}\ \text{C}$
3. $2.83 \times 10^5\ \text{C}$
4. $6.68 \times 10^7\ \text{C}$

Options 1. 1

2. 2

3. 3

4. 4

Q.26 Two coherent monochromatic light beams of intensities I and $16I$, respectively, superimpose over each other. The resultant intensity in the constructive interference is

1. $16I$
2. $25I$
3. $9I$
4. $15I$

Options

1. 1
2. 2
3. 3
4. 4

Q.27 A capacitor of capacitance C is first charged by a battery of V volts. It is disconnected from the battery and is filled with a dielectric of dielectric constant K . The ratio of energy stored in the two cases, respectively will be

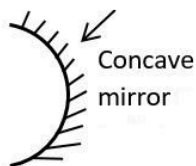
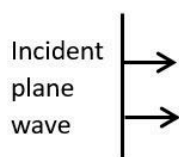
1. $K : 1$
2. $K^2 : 1$
3. $\sqrt{K} : 1$
4. $1 : K$

Options

1. 1
2. 2
3. 3
4. 4

Q.28

A plane wave is incident on a concave mirror of radius R . The shape of the reflected wavefront will be



1. diverging spherical
2. plane wavefront
3. converging spherical
4. cylindrical

Options

1. 1
2. 2
3. 3
4. 4

Q.29 A convex lens of focal length 40 cm is placed in contact with a concave lens of focal length 20 cm. If an object is placed at 40 cm in front of the combination, the position of the image formed by the combination is

1. 10 cm, on the same side of object
2. 20 cm, on the same side of object
3. 40 cm, on the other side of object
4. 20 cm, on the other side of object

Options

1. 1
2. 2
3. 3
4. 4

Q.30 A system consists of two protons and one electron. They are separated by a distance of 1 Å from each other. If the potential energy at infinity is taken to be zero, the potential energy of the system is

1. -13.6 eV
2. -14.4 eV
3. -16.2 eV
4. -19.2 eV

Options

1. 1
2. 2
3. 3

Q.31 The ratio of lengths of two wires is 2 : 3 and the ratio of their resistivities is 3 : 2. If they have the same resistance then the ratio of their respective diameters is

1. 4 : 9
2. 9 : 4
3. $\sqrt{2} : \sqrt{3}$
4. 1 : 1

Options

1. 1
2. 2
3. 3
4. 4

Q.32 Which of the following is NOT the property of electric field lines?

1. They start from positive charges and end at negative charges.
2. They can never cross each other.
3. They form closed loops.
4. They are continuous curves in a charge-free region.

Options

1. 1
2. 2
3. 3
4. 4

Q.33 The phase angle between the voltage and the current in an LR series circuit having the same resistance as its reactance is

1. 180°
2. 90°
3. 60°
4. 45°

Options

1. 1
2. 2
3. 3
4. 4

Q.34 A storage battery of emf 8.0 V and internal resistance 0.5Ω is being charged by a 60 V dc supply using a series resistor of 12.5Ω . The terminal potential difference between the terminals of the battery during charging is

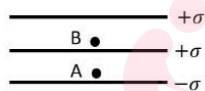
1. 5.4 V
2. 6.0 V
3. 10.0 V
4. 10.6 V

Options

1. 1
2. 2
3. 3
4. 4

Q.35

In this diagram, three infinite plane sheets of charges are shown with surface charge densities $+\sigma$, $+\sigma$, $-\sigma$. The value of electric field at points A and B are



1. 0 at A and 0 at B
2. $\frac{\sigma}{2\epsilon_0}$ at A and $\frac{3\sigma}{2\epsilon_0}$ at B
3. $\frac{3\sigma}{2\epsilon_0}$ at A and $\frac{\sigma}{2\epsilon_0}$ at B
4. $\frac{3\sigma}{2\epsilon_0}$ at A and 0 at B

Options

1. 1
2. 2
3. 3
4. 4

Q.36 Carbon, silicon and germanium have four valence electrons each. Their valence and conduction bands are separated by

energy gap E_g . If we represent $(E_g)_C$, $(E_g)_{Si}$ and $(E_g)_{Ge}$ as their respective energy gaps, then which of the following is true

1. $(E_g)_C = (E_g)_{Si} = (E_g)_{Ge}$
2. $(E_g)_C > (E_g)_{Si} > (E_g)_{Ge}$
3. $(E_g)_C < (E_g)_{Si} < (E_g)_{Ge}$
4. $(E_g)_C > (E_g)_{Ge} > (E_g)_{Si}$

Options

1. 1

2. 2
3. 3
4. 4

Q.37 A proton is accelerated by a potential V and the de-Broglie wavelength associated with it is λ . By how much potential a deuteron should be accelerated to acquire the same de-Broglie wavelength as that of a proton?

1. $2V$
2. $V/2$
3. $\sqrt{2}V$
4. $V\sqrt{2}$

Options

1. 1

2. 2
3. 3
4. 4

Q.38 The focal length of the objective of a compound microscope is f_o and its distance from the eyepiece is L . An object is placed at a distance u from the objective. For proper working of the instrument

- (A) $L < u$
- (B) $L > u$
- (C) $f_o < L < 2f_o$
- (D) $L > 2f_o$

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (B) and (D) only
3. (C) and (D) only
4. (B) and (C) only

Options

1. 1

2. 2
3. 3
4. 4

Q.39 A current is uniformly distributed across the cross-section of cylindrical wire of radius $R = 4$ mm with current density

$J = 2.5 \times 10^5 \text{ Am}^{-2}$. The amount of current flowing through the outer portion of the wire between radial distance $R/2$ and R is

1. 9.18 A
2. 94.2 A
3. 9.42 A
4. 7.28 A

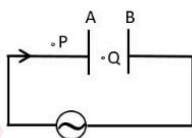
Options

1. 1

2. 2
3. 3
4. 4

Q.40

In the circuit shown below- a capacitor is being used in an AC circuit. A magnetic needle shows deflection, both at point P and Q. Which of the options among the following correctly specifies the reason for the presence of magnetic field at both the points?



1. Due to conduction current and displacement current at Q and P respectively .
2. Flow of charges in the wire and constant electric field between the plates A and B.
3. Flow of charges in the wire and varying electric field between capacitor plates A and B.
4. At Q the magnetic needle will show deflection due to magnetic field of the earth and at P due to the current flowing in the wire.

Options 1. 1

2. 2

3. 3

4. 4

Q.41 Positively charged particles are projected into a magnetic field. If the direction of magnetic field is along the direction of motion of the charged particles, the particles will experience

1. acceleration
2. deacceleration
3. deflection from the path
4. no change in velocity

Options 1. 1

2. 2

3. 3

4. 4

Q.42 Match List-I with List-II

List-I	List-II
Formula	Physical quantity

(A) $neAv_d$ (I) Mobility

(B) $\frac{v_d}{E}$ (II) Current

(C) $\frac{ne^2\tau}{m}$ (III) Conductivity

(D) $\frac{m}{ne^2\tau}$ (IV) Resistivity

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
3. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
4. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.43 If the A.C. main supply is given to be 220 V in a circuit. The average emf during a positive half cycle is

1. 52 V
2. 162 V
3. 198 V
4. 311 V

Options 1. 1

2. 2

3. 3

4. 4

Q.44 Which of the following statement is correct regarding a p-n junction diode.

1. During reverse bias, the width of the depletion region decreases.
2. During forward bias, the barrier height increases due to applied voltage.
3. The diode current during reverse bias is not much dependent on the applied voltage upto the breakdown voltage.
4. A milliammeter is used to measure current in a reverse biased diode.

Options 1. 1

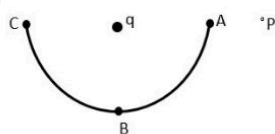
2. 2

3. 3

4. 4

Q.45

Consider the figure as shown below. If work done in bringing a point charge from infinity to points A, B and C are W_A , W_B and W_C respectively, then, which of the following relation is correct?



1. $W_A < W_B < W_C$
2. $W_A > W_B > W_C$
3. $W_A = W_B = W_C$
4. $W_A > W_B < W_C$

Options 1. 1

2. 2

3. 3

4. 4

Q.46 The slope of the stopping potential V_0 versus frequency ν of the incident radiation plot is (Where symbols have their usual meaning)

1. e/h
2. h/c
3. h/e
4. c/h

Options 1. 1

2. 2

3. 3

4. 4

Q.47 The first diffraction minima due to a single slit diffraction is at $\theta = 30^\circ$ for a light of wavelength $5000 \text{ \AA}$. The width of the slit is

1. $1.5 \times 10^{-5} \text{ cm}$
2. $1 \times 10^{-4} \text{ cm}$
3. $2.5 \times 10^{-5} \text{ cm}$
4. $1.25 \times 10^{-5} \text{ cm}$

Options 1. 1

2. 2

3. 3

4. 4

Q.48 A 25 cm long solenoid with 100 turns carries a current of 3 A. The flux density at the center of the solenoid is

1. $3.2 \times 10^{-2} \text{ Wb/m}^2$
2. $4.8 \times 10^{-3} \text{ Wb/m}^2$
3. $1.51 \times 10^{-3} \text{ Wb/m}^2$
4. $6.61 \times 10^{-2} \text{ Wb/m}^2$

Options 1. 1

2. 2

3. 3

4. 4

Q.49 A photon has a momentum of $3.3 \times 10^{-29} \text{ kg m/s}$. The frequency of the waves associated with the photon will be (Given: $h = 6.6 \times 10^{-34} \text{ Js}$)

1. $1.3 \times 10^3 \text{ Hz}$
2. $2.6 \times 10^3 \text{ Hz}$
3. $7.5 \times 10^{12} \text{ Hz}$

4. $1.5 \times 10^{13} \text{ Hz}$

Options

1. 1
2. 2
3. 3
4. 4

Q.50

The atomic mass of $^{12}_6\text{C}$ is 12.000000 u and that of $^{13}_6\text{C}$ is 13.003354 u. The energy required to remove a neutron from $^{13}_6\text{C}$ is

(Given: Mass of neutron is 1.008665 u and $1 \text{ u} = 931 \text{ MeV}$)

1. 2.73 MeV
2. 4.94 MeV
3. 6.08 MeV
4. 7.31 MeV

Options

1. 1
2. 2
3. 3
4. 4